MATHS MANTHAN
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 08 (QUADRATIC EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 13(iii).
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions (VSAQ) of 2 marks each.
4. **Section C** comprises 2 Short Answer questions (SA) of 3 marks each.
5. **Section D** comprises 2 Long Answer questions (LA) of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A (Multiple Choice & Assertion-Reason Questions 6 × 1 = 6 Marks)

Q1. If the quadratic equation $x^2 - kx + 9 = 0$ has equal roots, then the value of k is: [1]
(A) ± 3 (B) ± 6 (C) 6 only (D) 9

Q2. The number of real roots of the quadratic equation $(x-1)^2 - 4 = 0$ is: [1]
(A) 1 (B) 2 (C) 0 (D) 3

Q3. A quadratic equation whose roots are $(3+\sqrt{2})$ and $(3-\sqrt{2})$ is: [1]
(A) $x^2 - 6x + 7 = 0$ (B) $x^2 + 6x + 7 = 0$ (C) $x^2 - 6x - 7 = 0$ (D) $x^2 + 3x - 2 = 0$

Q4. If $ax^2 + bx + c = 0$ has equal roots, then the value of each equal root is: [1]
(A) $-\frac{b}{a}$ (B) $-\frac{b}{2a}$ (C) $\frac{b}{2a}$ (D) $\frac{c}{a}$

Q5. Which of the following quadratic equations has no real roots? [1]
(A) $x^2 - 4x + 3\sqrt{2} = 0$ (B) $x^2 + 4x - 3\sqrt{2} = 0$ (C) $x^2 - 4x - 3\sqrt{2} = 0$ (D) $3x^2 + 4\sqrt{3}x + 4 = 0$

Q6. Assertion-Reason: [1]
Directions: (A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.
Assertion (A): If 2 is a root of the equation $x^2 + bx + 12 = 0$, then $b = -8$.
Reason (R): A real number α is a root of the quadratic equation $p(x) = 0$ if and only if $p(\alpha) = 0$.

SECTION B (Very Short Answer Questions 2 × 2 = 4 Marks)

Q7. Find the value of k for which the equation $(k-4)x^2 + 2(k-4)x + 2 = 0$ has equal roots. [2]

Q8. Solve the following quadratic equation for x by factorization: [2]

$$2x^2 + x - 6 = 0$$

SECTION C (Short Answer Questions $2 \times 3 = 6$ Marks)

Q9. Solve for x: [3]

$$3\left(\frac{3x-1}{2x+3}\right) - 2\left(\frac{2x+3}{3x-1}\right) = 5, \quad x \neq -\frac{3}{2}, \frac{1}{3}$$

Q10. Seven years ago, Varun's age was five times the square of Swati's age. Three years hence, Swati's age will be two-fifths of Varun's age. Determine their present ages. [3]

SECTION D (Long Answer / HOTS Questions $2 \times 5 = 10$ Marks)

Q11. A motorboat whose speed is 15 km/h in still water goes 30 km downstream and returns back to the starting dock in a total time of 4 hours 30 minutes. Determine the speed of the river stream. [5]

Q12. An aircraft departed 50 minutes later than its scheduled departure time. In order to reach its destination 1250 km away on schedule, the pilot had to increase the speed by 250 km/h from its usual speed. Determine the scheduled speed of the aircraft and the original flight duration. [5]

SECTION E (Applied Case Study Problem $1 \times 4 = 4$ Marks)

Q13. Case Study: Municipal Swimming Pool & Dual Water Inflow Dynamics

The municipal sports authority is commissioning an Olympic-standard training pool. Two automated intake pipes are installed to fill the empty reservoir. Working together, both pipes fill the pool in $9\frac{3}{8}$ hours. The pipe of larger bore diameter takes 10 hours less than the pipe of smaller bore diameter to fill the pool when operating independently.

(i) If the smaller pipe takes x hours alone, write an expression for the portion of the pool filled by both pipes together in 1 hour. [1]

(ii) Formulate the quadratic equation in x in standard form modeling the filling process. [1]

(iii) Find the time required by each pipe separately to fill the pool. [2]

OR

Determine what fraction of the pool remains empty if both pipes operate simultaneously for only 5 hours.

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